DECODELABS

Project 3: SQL Data Analysis

PostgreSQL | E-commerce Orders Dataset

Project objective

Use structured SQL queries to retrieve, filter, group, aggregate, and sort order data to produce clear business summaries.

Dataset: 1,200 order records | 11 analysis-ready fields | Date range: 1 Jan 2023 – 30 Jun 2025

1. Project overview

This project uses PostgreSQL to analyse a cleaned e-commerce orders dataset. The work demonstrates the core SQL requirements in the training brief: SELECT, WHERE, ORDER BY, GROUP BY, COUNT(), SUM(), and AVG(). Additional queries use HAVING and date functions to extend the analysis.

2. Dataset and preparation

The supplied workbook contains 1,200 order rows. An import-ready CSV was created with 11 fields: order_id, order_date, product, quantity, unit_price, payment_method, order_status, items_in_cart, coupon_code, referral_source, and total_price. The cleaned coupon field was retained, with missing coupon entries represented as “No Coupon”.

Shipping addresses, tracking numbers, and customer IDs were excluded from the SQL extract because they are not needed for the required analysis. This reduces exposure of potentially sensitive identifiers while keeping the business analysis fields.

3. SQL methods used

SQL feature	Purpose in this project
SELECT	Choose the columns to display
WHERE	Filter individual order rows, such as Delivered orders
ORDER BY	Sort query results, for example highest order values first
GROUP BY	Create summaries by product, status, payment method, and month
COUNT()	Count orders in each group
SUM()	Calculate units and recorded order value
AVG()	Calculate average order value and average quantity
HAVING	Filter grouped results after aggregation
DATE_TRUNC / EXTRACT	Summarize orders by month and year

4. Descriptive findings

Measure	Validated result
Number of orders	1,200
Distinct customers in original workbook	1,189 (IDs excluded from SQL CSV)
Total recorded order value	1,264,761.96
Mean order value	1,053.97
Median order value	823.62
Total quantity ordered	3,535 units
Mean quantity per order	2.95
Date range	1 Jan 2023 – 30 Jun 2025
Order ID duplicates	0

5. Product analysis

Printer had the highest order count (181), followed by Tablet (179) and Chair (178). Phone had the lowest order count (156). Chair and Printer had the largest recorded order values in the product summary, at approximately 195,620.11 and 195,612.61 respectively. These are recorded order values across all statuses, not confirmed sales revenue.

Product	Orders	Units	Avg order value
Printer	181	542	1,080.73
Tablet	179	497	1,042.28
Chair	178	562	1,098.99
Laptop	173	535	1,110.56
Desk	170	508	985.06
Monitor	163	480	1,077.62
Phone	156	411	972.58

6. Order status analysis

Cancelled orders numbered 250 and Returned orders numbered 247. Together, these statuses account for 497 of 1,200 orders, or 41.42%. This is a notable operational pattern that a business team may wish to investigate. It does not, by itself, explain why orders were cancelled or returned.

Status	Orders	Share
Cancelled	250	20.83%
Returned	247	20.58%
Pending	237	19.75%
Shipped	235	19.58%
Delivered	231	19.25%

7. Time-based analysis

The dataset contains 510 orders in 2023, 459 in 2024, and 231 in January–June 2025. The 2024 count is lower than 2023, but 2025 covers only six months and must not be compared as a complete year. At monthly level, June 2024 had the highest count (53), while January 2025 had the lowest (27).

Year	Orders	Recorded order value	Average order value
2023	510	552,643.24	1,083.61
2024	459	480,235.87	1,046.27
2025	231	231,882.85	1,003.82

8. Outlier check

Using the IQR method on total_price: Q1 = 410.52, Q3 = 1,578.47, and IQR = 1,167.95. The upper fence is Q3 + 1.5 × IQR = 3,330.41. There are 8 order values above this upper fence. These values should be reviewed, not automatically deleted: they may represent legitimate high-value orders.

9. Key observations and recommendations

- Review the reasons behind cancelled and returned orders, because together they represent 41.42% of all records.
- Compare product order counts and average order values to understand differences in demand and order size.
- Review monthly order counts over a consistent period before drawing conclusions about seasonality.
- Investigate the eight high-value outliers to confirm whether they are valid orders or data issues.
- Define which order statuses should count as revenue before reporting sales or financial performance.

10. Limitations and interpretation

The sum of total_price includes every status: Cancelled, Returned, Pending, Shipped, and Delivered. Therefore, it is described as total recorded order value, not revenue or completed sales. The dataset does not explain the causes of order statuses, customer behaviour, or marketing effectiveness. SQL summaries describe patterns in the available records; they do not establish causation.

11. Files and execution

The accompanying SQL script creates the orders table and contains queries for all required SQL clauses and aggregates, plus optional analysis and data-quality checks. Import the CSV through pgAdmin 4's Import/Export Data dialog after creating the table. The numerical findings in this report were validated directly against the supplied workbook; the SQL still needs to be run in the user's local PostgreSQL environment to produce database execution screenshots or logs.

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